

Hierarchical Certified MPPI for Terrain-aware Mode-switching and Control. J. Song¹, V. Griffo², A. Gandhi², and Y. K. Nakka¹, ¹Daniel Guggenheim School of Aerospace Engineering, ²George W. Woodruff School of Mechanical Engineering, Georgia Institute of Technology. (Contact: willsong@gatech.edu)

Introduction: Sustained lunar operations necessitate rovers that are efficient over long traverses and capable across varied terrains. Wheeled platforms achieve low mechanical cost of transport (COT) but sink and slip in loose regolith [1]. Legged systems can traverse a wider range of terrains but sink under high contact forces on compressible soil [2]. Wheeled-legged hybrids can drive at low COT [3] while adapting to the terrain through different gait modes [4]. Existing research focuses on mode-switching based on energy criteria [5,6]; yet on the lunar surface, terramechanical variables—such as bearing capacity, compaction, and slip—dictate traversability but are not reliably inferred from cameras or LiDAR. We propose Hierarchical Certified MPPI, a two-level controller with a low-rate supervisor that estimates terrain conditions from proprioception and selects an appropriate locomotion mode, and a high-rate contraction-certified MPPI that executes it with a bounded tracking guarantee. We implement this on Multimodal Mini, a wheeled-legged rover test-bed.

Terrain-aware Supervisor: The supervisor takes a local elevation map and proprioceptive data to select from a library of locomotion modes including differential drive, high or low walking gaits, and an extrication behavior for recovery from immobilization. We build a dataset by repeating each mode over a range of terrain types, labeled with measured outcomes such as mechanical COT, slip ratio, and immobilization events. Because these outcomes are not predictable from terrain geometry alone, we train a classifier over the mode library from a fixed-length window of joint encoder, motor current, slip ratio, and local map features. Unlike prior proprioceptive methods, which demonstrate gait switching without formal guarantees [7], the supervisor passes its terrain estimate alongside the mode selection, preserving the executor’s guarantees.

Certified Executor: The selected mode is executed by a sampling-based control approach. Because model predictive path integral (MPPI) control evaluates candidate controls by simulating them forward rather than differentiating the dynamics [8], it can handle the abrupt change in model and cost that a switch causes. A contraction certificate also maintains sampling efficiency under

this switching and provides stability and probabilistic safety guarantees. Each mode therefore supplies its own dynamics model, executed under the terrain conditions estimated by the supervisor.

Validation & Plan: We demonstrate the locomotion modes on Multimodal Mini in a lunar-terrain environment through two scenarios, with mode-switching commanded open-loop. First, the robot traverses granular media of varying particle size at controlled slope angles, with motion capture providing ground truth. In Fig. 1, a high walking gait is used on a slope of fine sand and differential drive on flat ground.

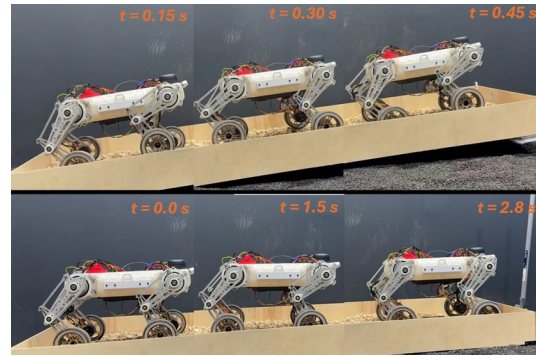


Fig. 1

In the second scenario, the robot’s wheel becomes stuck in loose gravel and a walking gait is deployed to resume its path, as shown in Fig. 2.

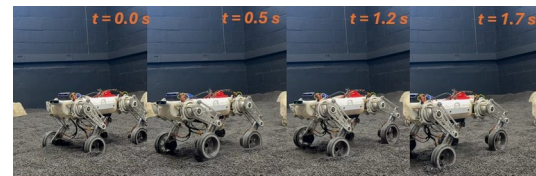


Fig. 2

Closed-loop mode-switching with the terrain-aware supervisor is the next step in the project.

References: [1] Callas J. L. (2015) *JPL Publication 16-2, JPL D-92756*, 31 pp. [2] Kolvenbach H. et al. (2018) *i-SAIRAS 2018*, ESA Conference Bureau. [3] Bjelonic M. et al. (2019) *IEEE Robot. Autom. Lett.*, 4, 2116–2123. [4] Lee J. et al. (2024) *Sci. Robot.*, 9, eadi9641. [5] Bjelonic M. et al. (2021) *IROS 2021*, 8388–8395. [6] Wang J. and Davies K. (2024) *Simul. Modelling Pract. Theory*, 132, 102893. [7] Elnoor M. et al. (2024) *IEEE Robot. Autom. Lett.*, 9, 7190–7197. [8] Williams G. et al. (2017) *ICRA 2017*, 1714–1721.